

Introduction

Anime is a style of animation that originated in Japan and has grown into a major global form of entertainment, appealing to audiences ranging from children to adults. In recent years, anime has become popular worldwide through streaming platforms, online communities, and social media. I have always enjoyed watching anime since middle school, where I watched many different series across a variety of genres. This interest led me to wonder what makes certain anime highly rated and how audience characteristics may influence viewing preferences.

Due to data availability, the analysis focused on datasets compiled in 2018, providing a snapshot of anime trends and user behavior during that period. However, a 2023 anime and manga dataset was also used to assess the correlation between manga scores and the scores of their anime adaptations.

This problem is important because it can reveal patterns in viewer behavior and preferences, while providing insights into how source materials are associated with anime success. It also helps examine whether demographic labels such as *shounen* and *shoujo*, which are intended to represent target audiences, are reflected in the viewing habits of male and female users. Understanding these relationships can help explain what factors contribute to highly rated anime and whether viewing habits differ across audience groups.

Dataset

This project used five datasets: AnimeList.csv, UserList.csv, and UserAnimeList.csv, which were compiled in 2018, along with anime.csv and manga.csv, which were compiled in 2023. The AnimeList.csv, UserList.csv, and UserAnimeList.csv datasets are part of the *MyAnimeList Dataset* published on Kaggle by Matěj Račinský (2018). The anime.csv and manga.csv datasets are part of the *MyAnimeList Anime and Manga Datasets* published on Kaggle by Andreu Vall Hernández (2023).

Table 1.

Dataset Name	Records	Columns	Key Variables
AnimeList.csv	14,478 anime entries	31	anime_id, title, type, source, score
UserList.csv	302,675 user records	17	username, gender
UserAnimeList.csv	80,076,112 interaction records	11	username, anime_id
anime.csv	24,985 anime entries	39	anime_id, title, type, source, score, demographics
manga.csv	64,833 manga entries	30	title, type, score

The datasets were collected by crawling information from MyAnimeList.net, an online platform where users track, rate, and review anime. Table 1 summarizes the structure and key characteristics of the anime and manga datasets. These datasets are appropriate because they contain user behavior data and anime metadata needed for analysis.

For the AnimeList dataset, only TV anime were selected, and the top 50 by score were retained using the title, score, and source columns. The anime.csv dataset was filtered to remove missing scores and restricted to TV anime adapted from manga, retaining only title and score. The manga.csv dataset was similarly cleaned by removing missing scores and keeping only manga entries with title and score columns. The anime and manga datasets were then merged on title to compare scores between manga and their anime adaptations.

For user preference analysis, the anime dataset was reduced to anime_id, title, and demographics, with missing values removed. Binary indicators for is_shounen and is_shoujo were created from demographic labels. The UserList dataset was filtered to include only Male and Female users, retaining username and gender. The UserAnimeList dataset was then restricted to users present in the cleaned UserList, keeping only username and anime_id.

The cleaned UserAnimeList dataset was merged with the anime demographics dataset using anime_id, adding the is_shounen and is_shoujo binary indicators to each anime based on its demographic classification. Missing values were set to 0. For each user, the proportion of shounen and shoujo anime watched were computed, producing shounen_ratio and shoujo_ratio. The ratios were defined as: $\text{shounen_ratio} = \frac{\text{sum}(is_shounen)}{\text{total anime user watched}}$, and $\text{shoujo_ratio} = \frac{\text{sum}(is_shoujo)}{\text{total anime user watched}}$. Finally, this dataset was merged with the cleaned UserList dataset on username to enable analysis of viewing preferences by gender.

One challenge I encountered was feature engineering the shounen and shoujo ratios. It required multiple preprocessing and merging steps to ensure that demographic labels were correctly mapped to each user's viewing history and accurately aggregated.

Research Questions

1. What are the top 50 TV Anime in 2018 based on score?
2. What is the distribution of source material among the top 50 TV Anime in 2018?
3. Is there a correlation between a manga's score and the score of its TV anime adaptation?
4. Do male and female users differ in their proportion of shounen anime in their viewing history?
 - **H₀ (null):** Male and female users have the same mean shounen_ratio.
 - **H₁ (alternative):** Male and female users differ in their mean shounen_ratio.
5. Do male and female users differ in their proportion of shoujo anime in their viewing history?
 - **H₀ (null):** Male and female users have the same mean shoujo_ratio.
 - **H₁ (alternative):** Male and female users differ in their mean shoujo_ratio.

Methods

Several statistical techniques were used to answer the research questions, including regression, correlation, hypothesis testing, and effect size measures. These methods were appropriate because the project examined relationships between variables and differences between user groups. To identify the top 50 TV anime in 2018 and the distribution of their source materials, frequency counts were used. This was appropriate for summarizing the characteristics of highly rated anime.

To examine the relationship between manga scores and anime adaptation scores, linear regression was performed using manga_score as the predictor and anime_score as the response variable. Pearson correlation was also calculated to measure the strength and direction of the relationship. In addition, bootstrap resampling was used to estimate 95% confidence intervals for the regression slope and correlation coefficient. These methods were appropriate for analyzing associations between two continuous variables.

To test whether male and female users differed in the proportions of shounen and shoujo anime they watched, I calculated the observed difference in mean viewing ratios and performed a permutation test by shuffling gender labels to generate a null distribution. Welch’s t-tests were also used to compare group means while allowing for unequal variances. These methods were appropriate for comparing two independent groups. Cohen’s d was calculated to measure the practical size of any gender differences. This was useful because statistically significant results may still have small real-world effects.

Results and Analysis

Figure 1.

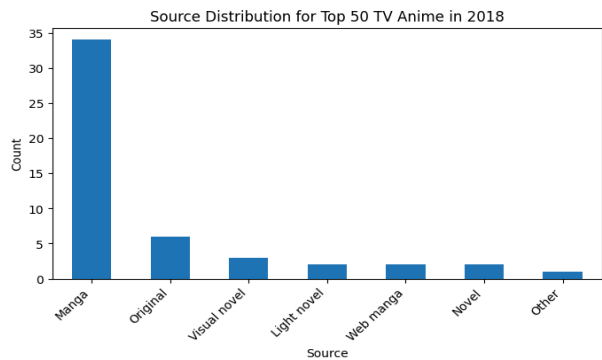


Figure 2.

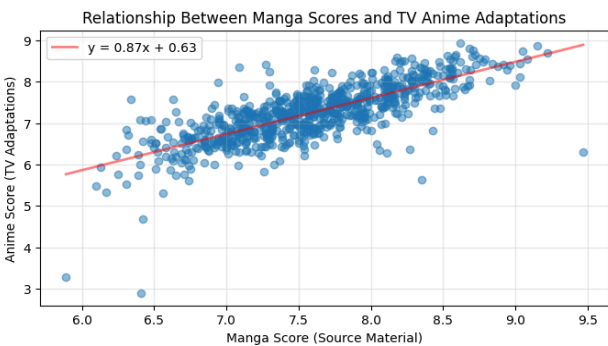


Figure 3.

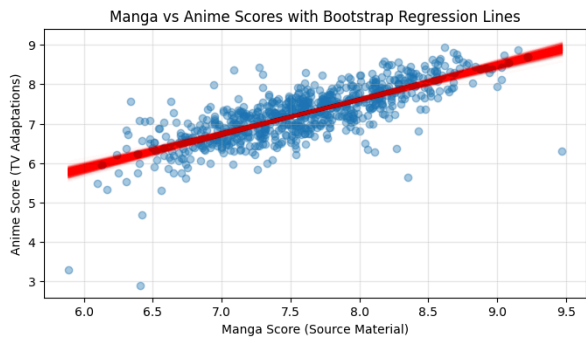


Figure 4.

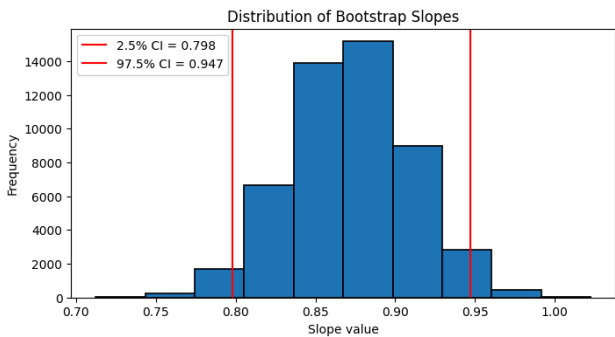


Figure 1 shows the distribution of source material among the top 50 TV anime in 2018. Manga was the most common source (34 titles), followed by original works (6) and visual novels (3). Overall, 88% were adaptations, suggesting that established readership may be associated with anime popularity. The complete table of the top 50 TV anime in 2018 can be found in the notebook. Figure 2 examines the relationship between manga scores and their TV anime adaptation scores. A positive linear trend is shown by the regression equation $y = 0.87x + 0.63$, meaning a one-point increase in manga score is associated with an expected 0.87 point increase in anime score. However, there are notable outliers indicating that manga score alone does not fully predict adaptation success. Figures 3 and 4 validate this relationship using bootstrap resampling. The regression slopes formed a narrow band around the original line, showing consistency across samples. The 95% confidence interval for the slope was $[0.798, 0.947]$, which lies entirely above zero, confirming a statistically reliable positive relationship.

Figure 5.

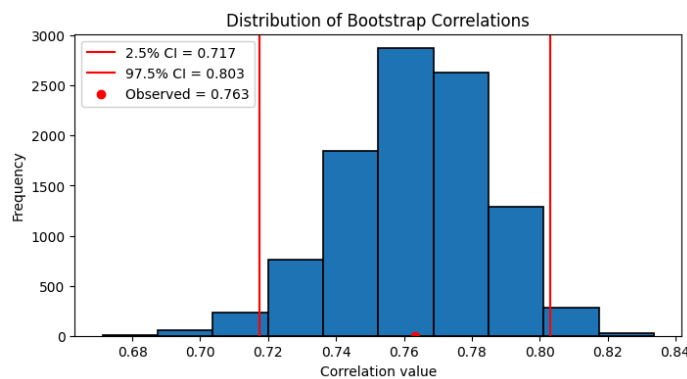


Figure 5 further supports this result. The observed Pearson correlation was $r = 0.763$, where $p < 0.001$, with a 95% bootstrap confidence interval of $[0.717, 0.803]$, indicating a strong and stable positive correlation across resamples. The model produced $R^2 = 0.583$, meaning manga score explains 58.3% of the variation in anime scores, while the remaining 41.7% is likely due to factors such as studio quality, budget, release timing, or directing choices.

Figure 6.

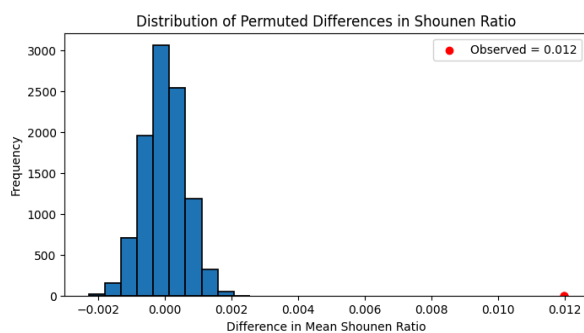


Figure 7.

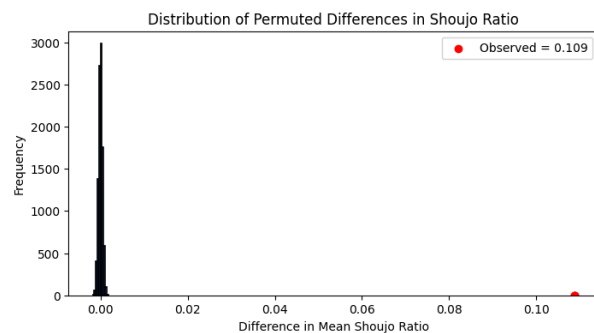


Figure 6 shows that the observed difference in shounen ratio (0.012) is statistically significant ($p < 0.001$), leading to rejection of the null hypothesis. Welch's t-test also confirms

this result ($t = 19.37, p < 0.001$), however, the effect size was negligible, as Cohen's $d = 0.088$ is below the conventional threshold of 0.2 for a small effect. This suggests the significance is largely due to the very large sample size rather than a meaningful practical difference. Figure 7 shows that the observed difference in shoujo ratio (0.109) is also statistically significant ($p < 0.001$), so the null hypothesis is rejected. Welch's t-test supports this finding ($t = 229.35, p < 0.001$), and Cohen's $d = 1.288$ exceeds the conventional threshold of 0.8, indicating a large and practically meaningful difference, with group means differing by more than one standard deviation.

Discussion

The results suggest that source material is a meaningful factor in anime adaptation success. Most of the top 50 TV anime in 2018 were adaptations, suggesting that pre-existing audiences may contribute to higher visibility and reception of the anime adaptations. A strong positive relationship was observed between manga and anime adaptation scores, with higher-rated manga tending to be associated with higher-rated adaptations. Bootstrap analysis confirmed that this relationship is stable and statistically reliable.

The user preference analysis found statistically significant gender differences in shounen and shoujo viewing ratios. However, the shounen effect size was negligible, indicating little practical difference, whereas the shoujo ratio showed a large effect size, suggesting a meaningful difference between groups. Overall, shounen consumption was similar across genders, while shoujo consumption varied more strongly, suggesting that shoujo viewing patterns differ more strongly by gender than shounen viewing patterns.

Some limitations of this study include the use of only 2018 and 2023 data, which limits generalizability across years, and the fact that 41.7% of the variation in anime scores remains unexplained, suggesting that factors such as studio, budget, and directing choices also play a role. Future work could incorporate additional variables such as studio and genre, expand the dataset across more years, and apply multivariate regression or machine learning methods to improve predictive accuracy.

Use of AI Tools

ChatGPT was used as a support tool for analysis guidance and writing assistance, including suggesting Cohen's d to complement hypothesis testing with effect size interpretation and helping condense paragraphs.

Conclusion

The study found that most top-rated 2018 TV anime were adaptations, mainly from manga, and that manga scores are positively correlated with anime scores, suggesting that source quality is an important factor in adaptation success. User analysis showed negligible gender differences in shounen consumption but a large difference in shoujo consumption, indicating stronger gendered variation in shoujo viewing preferences.

Acknowledgements

Andreu Vall Hernández, “MyAnimeList Anime and Manga Datasets.” Kaggle, 2023.
Available at: <https://www.kaggle.com/datasets/andreuvallhernandez/myanimelist/data>

Matěj Račinský, “MyAnimeList Dataset.” Kaggle, 2018. doi: 10.34740/KAGGLE/DSV/45582.
Available at: <https://www.kaggle.com/datasets/azathoth42/myanimelist/data>